

WAIFINDERS

What It Does — In Plain Language

For First Nations, Funders, and Partners · August 2026

8	Hazard types monitored 58	BC First Nations scored 67	Nuclear plants tracked 2,000
0.8818	Accuracy score wildfire detection 7.15x	Better than random at finding high-risk fires 10 yr	Readiness projection for Nations (SFAS) 260

Bridges

Tests

WAIFINDERS is a decision-support platform that watches multiple hazards at once — wildfires, earthquakes, floods, hurricanes, tsunamis, and nuclear facilities — and tells you which ones need attention right now. It is built for Nations, infrastructure managers, and emergency planners who need to see the whole picture without waiting for separate reports from separate agencies.

Everything WAIFINDERS does is based on official public data. It does not make up numbers, predict the future, or replace your own judgment. It is a tool — the decision always stays with you.

THE CORE QUESTION WAIFINDERS ANSWERS

What is threatening us right now, and how serious is it?

WAIFINDERS watches eight types of hazards simultaneously and gives each one a score from 0 to 1 — where 0 means calm and 1 means the worst conditions on record. It then combines all those scores into a single decision: what level of response does today's situation actually require?

The four response levels, in plain language:

NORMAL OPERATION	Everything looks routine. Continue normal monitoring and planning.
MONITOR	One or more hazards are elevated. Watch more closely. No immediate action required yet.
MITIGATION REQUIRED	Significant hazard exposure. Take protective action now — check resources, review evacuation plans, alert key contacts.
EMERGENCY RESPONSE	Critical conditions or multiple hazards hitting at once. Full emergency response. This is when everything else stops.

What triggers an upgrade from MONITOR to EMERGENCY?

When two or more hazards reach the MONITOR level at the same time, WAIFINDERS automatically escalates the overall status by one level. A flood and a hurricane arriving together is more dangerous than either alone — and the platform recognises that. This is called a compound event.

WHAT WAIFINDERS WATCHES

Each hazard domain uses official public data and a transparent scoring formula. No scores are invented or estimated without a data source.

Wildfire — Alberta

What it watches: Forest fire danger across Alberta

How it scores: Fire Weather Index (FWI), fire spread rate, community exposure, firefighting resources

Data source: Official Alberta fire records 2006–2025

What the numbers mean: Accuracy: 0.8818 out of 1.0. Finds real large fires 7× better than chance.

Wildfire — BC First Nations

What it watches: Fire danger near First Nation territories in BC

How it scores: Fire danger index, proximity to active fires, fire size, fire spread speed

Data source: BC Wildfire Service public records; BC Data Catalogue territory locations

What the numbers mean: 58 Nations scored. Kwadacha Nation highest (44 km from Donnie Creek 2023 fire).

Earthquake

What it watches: Ground shaking from earthquakes

How it scores: Earthquake magnitude, depth, and nearby population

Data source: USGS ShakeMap — the US Geological Survey's official earthquake maps

What the numbers mean: A full Cascadia megquake (M9.0) scores MITIGATION REQUIRED.

Tsunami

What it watches: Ocean waves from underwater earthquakes

How it scores: Wave height, earthquake size, how far inland the wave could reach

Data source: NOAA DART buoy network — ocean sensors across the Pacific

What the numbers mean: The 2011 Japan tsunami scores 0.970 — near the top of the scale.

Flood

What it watches: River and coastal flooding

How it scores: Water surge height, rainfall

Data source: USGS water gauge network; NOAA tide gauge readings

What the numbers mean: Hurricane Sandy's flood surge (2.74 m) scores MITIGATION REQUIRED.

Hurricane

What it watches: Tropical storm and hurricane winds

How it scores: Wind speed, storm surge, distance to coastline

Data source: NOAA's 170-year Atlantic hurricane database (1851–2023)

What the numbers mean: Hurricane Andrew 1992 scores 0.814 — EMERGENCY RESPONSE.

Nuclear

What it watches: Nuclear plant risk based on size, nearby population, and current power output

How it scores: Plant capacity, population in the emergency zone, current reactor power

Data source: US Nuclear Regulatory Commission daily power status report

What the numbers mean: 67 US plants scored. High-density plants in the Northeast score highest.

BC FIRST NATIONS — WHAT WAIFINDERS PROVIDES

What does a Nation actually see?

Each Nation gets a current fire threat score (WARN) and a 10-year readiness projection (SPARK7). The WARN score reflects how close the fires are today, how fast they are moving, and how large they are. The SPARK7 projection looks further — it asks: given all the pressures this community faces (heat, drought, storms, wildfire, sea level), how does their readiness look in 10 years?

What is SPARK7?

SPARK7 is a readiness model originally built for Indigenous communities worldwide. WAIFINDERS has now connected it to live BC fire data. A community with a readiness score above 55 is GREEN (resilient). Between 40–55 is ORANGE (watch). Below 40 is RED (priority). All 58 BC Nations currently score ORANGE — meaning fire pressure is already affecting their readiness outlook.

How does WAIFINDERS handle Nation data?

WAIFINDERS uses only public data: BC Wildfire Service fire records and approximate territory locations from the public BC Data Catalogue. No cultural data, sacred site information, governance data, or Nation-specific information is collected or used. Territory coordinates are approximate.

OCAP principles (Ownership, Control, Access, Possession) are built into every part of the BC module. The Nation owns its data. WAIFINDERS does not claim to represent any Nation's view of their situation, and no Nation has endorsed this platform. A formal data agreement with the Nation is required before territory-specific data can be used.

What would change with a Nation data agreement?

With a formal agreement, WAIFINDERS could:

- Use the Nation's own territory boundary maps instead of public approximations
- Score specific community infrastructure (roads, water systems, health centres)
- Track evacuation route risk under active fire conditions
- Provide the Nation with their own data — stored and controlled by them
- Integrate traditional ecological knowledge and seasonal fire patterns if the Nation chooses

INFRASTRUCTURE — WHAT IS AT RISK WHEN A HAZARD HITS

What does PULSE do?

PULSE scores infrastructure — bridges, roads, water mains, rail — on how stressed they are likely to be under current hazard conditions. A bridge that is already in poor condition AND sits in an earthquake zone AND a flood plain gets a much higher risk score than a newer bridge in a low-hazard area. PULSE shows you which assets need attention

before the hazard arrives.

Current results — 2,000 bridges in New York State:

Colour	Count	What it means
RED	200	Highest combined risk — poor condition in a high-hazard zone. Immediate priority.
AMBER	340	Elevated exposure. Monitor closely and prioritise inspection.
YELLOW	1,141	Moderate exposure. Schedule inspection in next cycle.
GREEN	319	Low combined risk. Routine maintenance cycle.

Hazard zone sources: [earthquake risk from the US Geological Survey 2014 national map](#); [tsunami exposure from NOAA coastal gauges](#); [flood risk from FEMA flood maps](#).

Has this been tested on real infrastructure?

Yes. PULSE was validated against real open-data records in Toronto (4,248 assets, accuracy 0.74) and Calgary (4,748 assets, accuracy 0.66). Both cities confirmed their open data can be used. The next step is working with a city or utility to use their live operational data instead of open data.

WHAT WAIFINDERS DOES NOT DO

Being clear about limits is as important as describing capabilities. The following are hard rules — not disclaimers.

Does not predict hazards

WAIFINDERS does not predict when or where a fire, earthquake, or flood will occur. It scores the severity of hazards that are already happening or already recorded.

Does not replace human judgment

Every score is a decision-support input. A person — a Nation leader, an emergency manager, an engineer — always makes the final call. WAIFINDERS does not direct action.

Does not use Nation data without consent

No cultural, spiritual, traditional territory, or governance data is collected from any Nation without a formal written agreement following OCAP principles.

Does not claim government endorsement

No BC Wildfire Service, NRC, USGS, NOAA, FEMA, or government body has endorsed WAIFINDERS. Data sources are cited; endorsement is not claimed.

Does not fabricate numbers

Every score comes from a named official data source. If a data source is unavailable, WAIFINDERS says so explicitly — it does not substitute made-up values.

Is not a production emergency system

WAIFINDERS is pilot-ready. It is built to the standard required for production, but it has not completed external peer review or received an official data agreement. Those steps are in progress.

WHAT IS READY TODAY

Capability	Ready for pilot?	What is still needed for production
Wildfire WARN — Alberta	YES	External peer review; official Alberta fire data agreement
Wildfire WARN — BC First Nations	YES	BC official fire data agreement; Nation data agreements
SPARK7 — Nation 10-year readiness	YES (BC connected)	Nation-validated baselines; formal data governance
Earthquake / Tsunami / Flood / Hurricane	YES	Regional calibration; external review
Nuclear plant tracking	YES	Scope agreement; utility partner
Infrastructure PULSE	YES	Production operational data partnership (city or utility)
WISE decision engine	YES	Pilot integration; knowledge corpus; external review
Audit trail (SENTINEL / S.A.F.E.)	YES	Federal partner engagement (underway)
Live Nation dashboard	YES	Nation data agreement; formal engagement

HOW TO ENGAGE

For First Nations

- Request a demonstration of the BC wildfire dashboard for your territory
- Begin a Nation-to-Nation conversation about a data agreement (OCAP framework)
- Identify a data steward within the Nation to oversee any future data sharing
- Define what questions the Nation wants WAIFINDERS to help answer

For funders and government partners

- Review the full technical build metrics (separate PDF — see enclosed)
- Request the SENTINEL federal engagement package
- Identify a pilot context: wildfire season, infrastructure assessment, emergency planning
- Commission an independent external review of the WARN model

For infrastructure and utility partners

- Share a sample of operational asset data (pipe condition, bridge inspection) for a pilot PULSE run
- Define the infrastructure question: what assets are you most worried about?
- Review the Toronto and Calgary validation results

WAIFINDERS is a decision-support platform — not a forecasting system, not an emergency directive system, and not a government-endorsed product. All wildfire performance figures are from a real replay of Alberta fire records (Open Alberta OGL-A licence, 2006–2025). Alberta scope only — performance does not transfer to BC or Ontario without a separate study. OCAP principles govern all engagement with Nations.